

OVERCOMING GRAVITY

SECOND EDITION

**A SYSTEMATIC APPROACH TO
GYMNASTICS AND BODYWEIGHT STRENGTH**

STEVEN LOW, DPT

Overcoming Gravity: *A Systematic Approach to Gymnastics and Bodyweight Strength*
Second Edition

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INTRODUCTION

I never imagined that *The Fundamentals of Bodyweight Strength Training*, the article I wrote for *Eat. Move. Improve.* in March of 2010 would become so popular. After publishing my first book, *Overcoming Gravity: A Systematic Approach to Gymnastics and Bodyweight Strength* in November of 2011, I was overwhelmed by how much support it received. I knew I had to release a second edition, because the first edition did not fully accomplish what I had set out to accomplish. My primary goal for the first edition was to create a comprehensive resource for beginners to learn how the human body works, and to aid the reader as they construct their own workout routines. You have certainly heard the axiom “Give a man a fish and feed him for a day; teach a man to fish and feed him for a lifetime.” I wanted to build the information base to feed the aspiring athlete for a lifetime. As is often the way with a work, once published, use of the book brought about many helpful questions. I published several articles on *Eat. Move. Improve.* to clarify content from the book. This edition—not unlike strength training—is built on examination of the successes and the aspects in need of focused work.

Since the release of the first edition, I have learned a lot about simplicity and teaching methods. I hope this second edition will guide you toward learning all you need to meet your own goals.

Mastering your body requires a lot of hard work and persistence, but the potential for overall results is astounding. Bodyweight exercises can be performed almost anywhere with minimal equipment, are fun to play around with, and it doesn’t hurt that they can be visually impressive. Strength levels acquired from proper bodyweight training transfers over to all other forms of strength, including weight training. Bodyweight strength training is thus extremely rewarding.

Nothing worthwhile comes without hard work and a fair share of frustration. Bodyweight strength training is no exception. Unlike with barbells there are very few gauges for progress. Athletes may become stuck on certain strength progressions for weeks or even months at a time with little clue how to push beyond plateaus. Stagnation is a very real problem, but with good programming we can fight this tendency. Programming is all about planning, and a good plan will help minimize the plateaus and keep us moving to our individual pinnacle.

Strength and conditioning has been constantly refined and modified in most of the major sports such as track and field, football, basketball, swimming, and many others. However, there are few people who know how to effectively implement progressions and programming in the context of bodyweight strength. Gymnastics gyms simply do not have the monetary resources or consumer demand to hire strength and conditioning professionals to help refine the physical preparation that is required. Likewise, there are no requirements for coaches to know much about strength and conditioning. Thus, there are limited sources of true bodyweight strength programming available. Most of the information is in the heads of high-level gymnastics coaches who do not have the time or inclination to record their expertise. Similarly, knowing progressions for exercises alone does not necessarily mean that one has enough knowledge to implement effective programming.

One such consideration is the importance of *populations*. The way you train a gymnast will not be the way you train a recreational adult interested in the same material.

Overcoming Gravity is an attempt to change that. The primary goal of this book is to allow beginner and intermediate athletes to delve into the world of bodyweight strength training and progress effectively and safely. This book will equip you with the knowledge to build safe and effective workouts and progress in your bodyweight strength movements. Additionally, there will be condensed exploration on subjects that relate to general movement such as endurance, metabolic conditioning, cardio, nutrition, and the like, to add support and stability to the programming and subsequent training.

To equip you with the right tools, I have distilled it in what I call the *fundamental knowledge base*. If you are a personal trainer or a coach this book will serve as a good knowledge base for working with your clients.

Fundamental Knowledge Base

The purpose of the fundamental knowledge base is threefold:

- To provide general training knowledge.
- To emphasize the differences between various populations.
- To combine general training knowledge and the differences in specific populations into examples of training routines at various levels of ability.

The information in the first two categories is broken down into distinct sections:

General Training Knowledge

- Physiological concepts behind strength and hypertrophy training.
- How to set the right goals.
- How to structure routines.
- The methods for programming/planning a routine.

Differences in Various Populations

Training for different skill levels—beginners, intermediates, advanced, and beyond. Training between the sedentary and the active. Training between the young and the old. Training between those who are using exercises for their sport and for those who train for other reasons. Considerations for those who are injured or recovering from an injury.

Section three will integrate the first two categories. If you are confused on the importance of specific information in a chapter, the chapter summaries will have each of these sections broken down into three main aspects. This will enable you to learn to apply concepts correctly within the framework of a routine.

- Knowledge Base
- Application